Cooper Kimball-Rhines, Ph.D.
Data Science, Genomics, Conservation

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I study how plants with limited genetic variation adapt to their environment. My research is collaborative and interdisciplinary with a focus on implementing genetic data in plant conservation planning.

Education

Ph.D., Environmental Biology, 2026
University of Massachusetts Boston
NSF Graduate Research Fellow
Dissertation: Adapting population genetics, practice, and policy for plant conservation in a changing world
Advisor: Brook Moyers, brook.moyers@umb.edu
Committee: Mayra C. Vidal, Georgia Mavrommati, Alex Harkess

B.Sc., Microbiology and Immunology, 2021
McGill University, School of Biomedical Sciences
Dean's Multidisciplinary Research List
Thesis: The PFAS package in *enviPath*: a resource for the prediction of AFFF biotransformation; Advisor: Jinxia Liu

Work Experience

Pathways Intern, United States Environmental Protection Agency, Oct. 2021-Aug. 2024
EPA Student Intern of the Year 2023
Region 1 Superfund and Emergency Management Division
Assisted on-scene coordinators in assessing contamination at emergency removals
Manager: Stacy Greendlinger, stacy.greendlinger@epa.gov

Roadside Pollinator Assistant, New Hampshire Fish and Game, May-Sept. 2019-2021
Performed monarch butterfly surveys to inform NH DOT policy on habitat management
Supported Karner Blue Butterfly, Frosted Elfin, and White Mountain Fritillary projects
Manager: Heidi Holman, heidi.l.holman@wildlife.nh.gov

Educational Experience

Curriculum Maintainer, The Carpentries, Intro to R Geospatial Data, May 2025-present
Instructor, iTCGA Computation for Cancer Genomics Workshop, June 2-20, 2025
Instructor, The Command Line for Genomics Workshop, July 15-17, 2024
Instructor, GIS in R for Casco Bay Workshop, Jan. 10-12, 2024
Certified Instructor, The Carpentries, Dec. 2023-present
Teaching Assistant, UMass Boston, Biostatistics and Experimental Design, Sept-Dec. 2023

Research Grants and Fellowships

2023-2028	NSF Graduate Research Fellowship	$159,000
2023-2025	Nantucket Biodiversity Initiative Climate Change and Island Resilience	$5,506
2023-2025	DOE Joint Genome Institute Community Science Program	RNA-seq Services
2023	UMB Transdisciplinary Dissertation Proposal Development Program	$3,000
2023	UMB Nancy Goranson Endowment Fund	$735
2022	New England Botanical Society Graduate Student Research Award	$1,500

Invited Workshops

ESIIL	Leadership Program and Innovation Summit on AI for Sustainability: Translating Environmental Data into Decisions, May 11-14, 2026, Boulder, CO
NCEAS	Environmental Data Science Summit: AI and Conservation Management, Feb. 4-6, 2025, Santa Barbara, CA
AIBS	Policy Communication Bootcamp and Congressional Visits Day, April 28-30, 2025, Washington, DC

Organized Workshops

NECASC	Climate Adapted Restoration Meeting, April 9, 2026, Online hosted by the Northeast Regional Invasive Species and Climate Change Network, Amherst, MA

Publications

6. **Kimball-Rhines, C.,** Mello E., Taveras Guzman, S., Moyers B. Divergence in genotype and methylation are decoupled and are not driven by nitrogen source in a salt marsh plant. In prep for *Evolutionary Applications*
5. **Kimball-Rhines, C.,** Taveras Guzman, S., Mavrommati, G., Moyers, B. Leaping a conservation genetics gap: A collaborative, multidisciplinary approach to restoration genetics in the pine barren plant *Lupinus perennis*. In prep for *Conservation Science and Practice*
4. Keller, AG., **Kimball-Rhines, C.,** Leng, D., McIntosh, T., Nelson, K., Tension between opportunities and risks of deep learning in social-ecological systems research. In review, *Ecology and Society*
3. **Kimball-Rhines, C.,** Moyers, B. Missing the forest for the trees: A review and case study of state conservation law. Accepted, *Oryx*.
2. Comerford, M.S., Nash, K., Germaine, A.S., Andrews, C., Barr, T.S., Dain, J., Gonclaves, A., Jones, A., **Kimball-Rhines, C.,** Maguire, W., Richards, A., Schneider, E., Wilkes, A. and Vidal, M.C. (2026), Quantitative synthesis of the effects of drought on community composition and species interactions in terrestrial ecosystems. Oikos e12026. https://doi.org/10.1002/oik.12026
1. **Cooper Kimball-Rhines,** Alice E Kaisla, Scarlet Taveras Guzman, Emily Mello, Karol Vanessa Rojas Ramirez, Alex Harkess, Brook Moyers, The annotated, chromosome-scale Salicornia depressa (American pickleweed) genome, Journal of Heredity, 2025; esaf062, https://doi.org/10.1093/jhered/esaf062

Presentations

10. Poster: **Kimball-Rhines C.,** Taveras Guzman, S., Moyers B. "Gene Pool or Puddle: When 'Local is Best' Meets Restoration Reality." Ecological Society of America 2025; August, 2025, Baltimore, MD
9. Poster: **Kimball-Rhines C.,** Moyers B. "Divergence and Diversity Decoupled: A Population Methylomic Approach." Harvard Plant Biology Initiative Meeting; May, 2025, Boston, MA.
8. Poster: **Kimball-Rhines C.** "Drowning to Diverse: A causal analysis of salt marsh sediment placement." Northeast Estuarine Research Society Meeting; April, 2025, Provincetown, MA.
7. Oral Presentation: **Kimball-Rhines C.,** Palmer A., Harkess A., Guzman-Taveras S., Mello E., Rojas V., Moyers B. "Six Hot Tips learned from assembling a tetraploid halophyte genome." Third Joint Congress on Evolutionary Biology; July, 2024, Montreal, QC, CA.
6. Oral Presentation: **Kimball-Rhines C.,** Palmer A., Moyers B. "Genetics for Marsh Conservation." Nantucket Biodiversity Initiative Meeting; November, 2023, Nantucket, MA.
5. Seminar: **Kimball-Rhines C.,** Palmer A., Moyers B. "Live Demonstration of DNA Sequencing for Marsh Conservation." Nantucket Field Station; August 15, 2023, Nantucket, MA.
4. Poster: **Kimball-Rhines C.,** Moyers B. "One Fish, Two Fish, Sea Fish, Bee Fish: Assessing State-Level Biodiversity Policy." Evolution in Small Population Meeting; July, 2023, Princeton, NJ.
3. Poster: **Kimball-Rhines C.,** Moyers B. "One Fish, Two Fish, Sea Fish, Bee Fish: Assessing State-Level Biodiversity Policy." Society for the Study of Evolution; June, 2023, Albuquerque, NM.
2. Poster: **Kimball-Rhines C.,** Palmer A., Moyers B. "Genome Profiling of Salicornia depressa." Ecological Society of America Meeting; August, 2023; Montreal, QC, CA.
1. Lightning Talk: **Kimball-Rhines C.,** "Genome Profiling of Salicornia depressa" UMass Boston Earth Day Symposium; March 2022; Boston, MA.

References

Dr. Brook Moyers, Assistant Professor, University of Massachusetts Boston
Ph.D. Advisor, brook.moyers@umb.edu

Stacy Greendlinger, Enforcement Coordinator, U.S. Environmental Protection Agency
Former Supervisor, stacy.greendlinger@epa.gov

Heidi Holman, Wildlife Biologist, New Hampshire Fish and Game
Collaborator and Former Supervisor, heidi.l.holman@wildlife.nh.gov